Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Java, Python, SQL, C/C++
Core Computer Science: DSA, OOP, Design Patterns, Problem Solving, Multithreading, Exception Handling
Generative AI & LLMs: Large Language Models (LLMs), Generative AI, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Google Gemini API, Context Engineering, LLM Application Development, Model Evaluation
Backend Development: FastAPI, Flask, REST APIs, API Integration, TCP, Request Handling, Authentication, JSON
Databases: MongoDB, MySQL, SQLite, Redis, Database Design, Query Optimisation
Software Engineering & Testing: Git, GitHub, PyTest, Unit Testing, Test-Driven Development (TDD)
Tools & Frameworks: React.js, Node.js, Scikit-learn, Pandas, NumPy

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***API Rate Limiter Library* |LINK**

- Built a Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API automation and reliable traffic control.
- Designed a modular, thread-safe architecture with pluggable storage backends to support concurrent request handling and reliable API behavior across application environments.
- Developed 30+ PyTest automated tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for reusable software development.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025